Jeppe Update — 2026-07-28

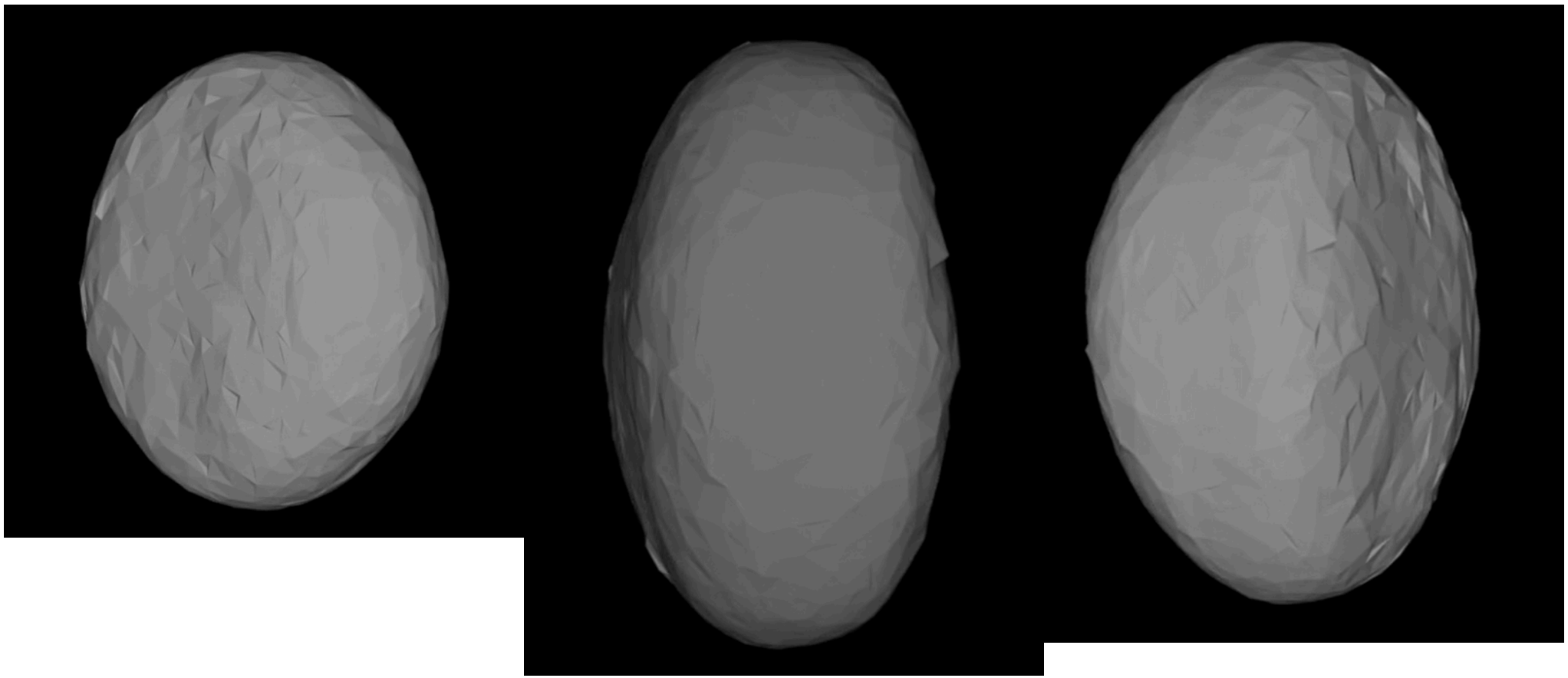


Figure 1: Ugly yet motivational asteroid 6 reconstruction.

Mitsuba Forward

- Code from last time cleaned up a bit and available in `hac.mitsuba`
  - Still a little messy
  - I may break quite a few things in the future, but usable by others for experiments for now.
- Overview of new modules:
  - `scene`: module for constructing batch Mitsuba scenes and rendering them
    - `debug_scene`: to make the nice plots shown later
  - `visualize`: module for visualizing meshes and light curves
  - `forward`: module for forward rendering of meshes and light curves
    - Differentiable, alledgedly also *PyTorch* compatible

Batch Scenes — z-stack

- To (efficiently) render + store gradients of mesh at multiple angles and *phases* (of the rotation), we set up a single scene
  - Each *phase* is an affine transformation on reference to the mesh
  - i.e. stacked and rotated along z
  - Each of these has sensors for enabled observation angles
- **Live Demo?**

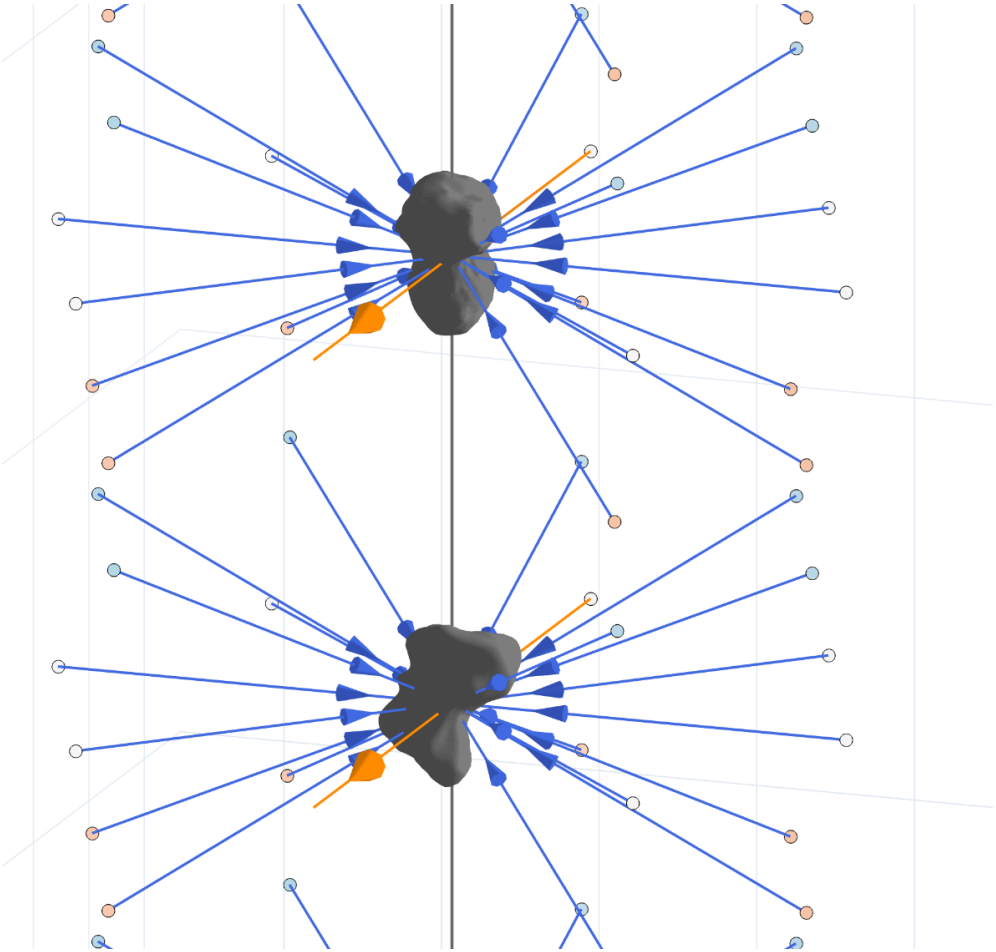


Figure 2: Batch scene setup for asteroid 3, see `mitsuba_z_stack_viewer.ipynb` for details.

Batch Scenes — Forward

- Forward rendering is then just rendering each sensor individually
- Then applying appropriate reduction to get light curve
- Note: Currently this is just sum without thresholding
  - Does not match *Otsu/binary* or *intensity* methods
  - Given no 'stray light', so maybe thresholding not needed to match *intensity* method well?
  - Tricky to get thresholding to be differentiable, but possible with *sigmoid* or other

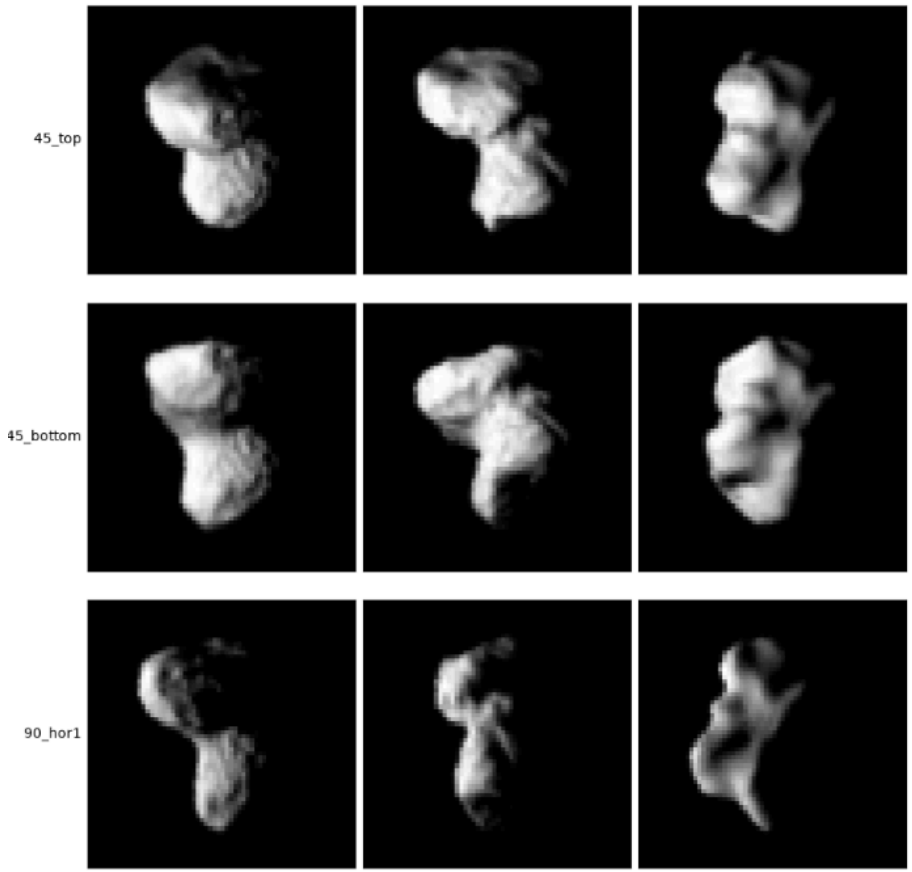


Figure 3: Subset of sensor renderings in a batch scene for asteroid 3 using Mitsuba forward model. Phases are $\phi \in \{0°, 90°, 180°\}$ across, respectively.

Optimization

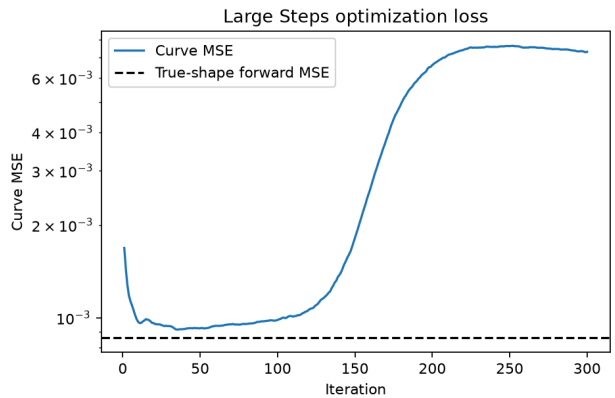
- Follows `https://mitsuba.readthedocs.io/en/stable/src/inverse_rendering/shape_optimization.html`
- See `workspaces/jeppe/mitsuba_shape_optimization.py` for details
- Still lots to do, very basic at the moment
- Scale penalty is kind of wrong
  - Maybe only penalize (relative) point mass outside bounding cylinder?
- Try on all models
- New ‘built-in’ penalty by differentiably applying normalization to mesh as part of forward
  - Seems to have helped with ‘pan-caking’

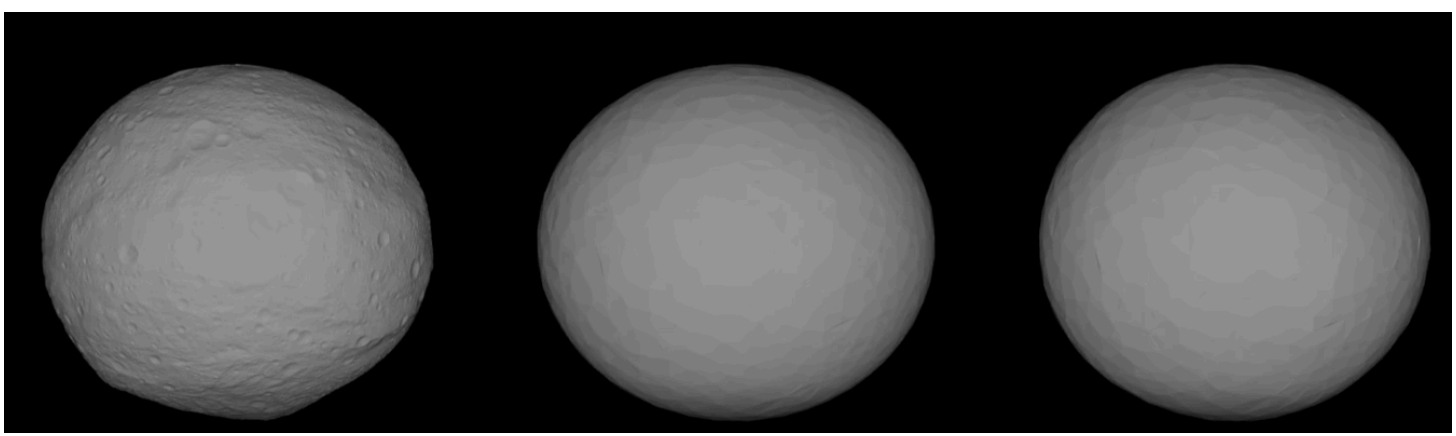
Optimization (Currently)

- Currently *relatively* fast
  - A10 GPU: ≈ 30 mins
  - A100 GPU: ≈ 20 mins
  - DrJIT needs Compute Capability 7.5+, so no V100
  - Priority access to A10/A40/A100 would be useful later
- Parameters currently:
  - Phases: $N_\phi = 180$
  - Faces: $N_f = 2500$
  - Samples per pixel: $N_{\text{spp}} = 8$
  - Resolution: $(H, W) = (64, 64)$
  - Iterations: $N_{\text{iter}} = 300$
  - Adam with LR $\alpha = 2 \times 10^{-2}$
  - “Large Steps” parameter, $\lambda = 25.0$

Reconstruction: Asteroid 1

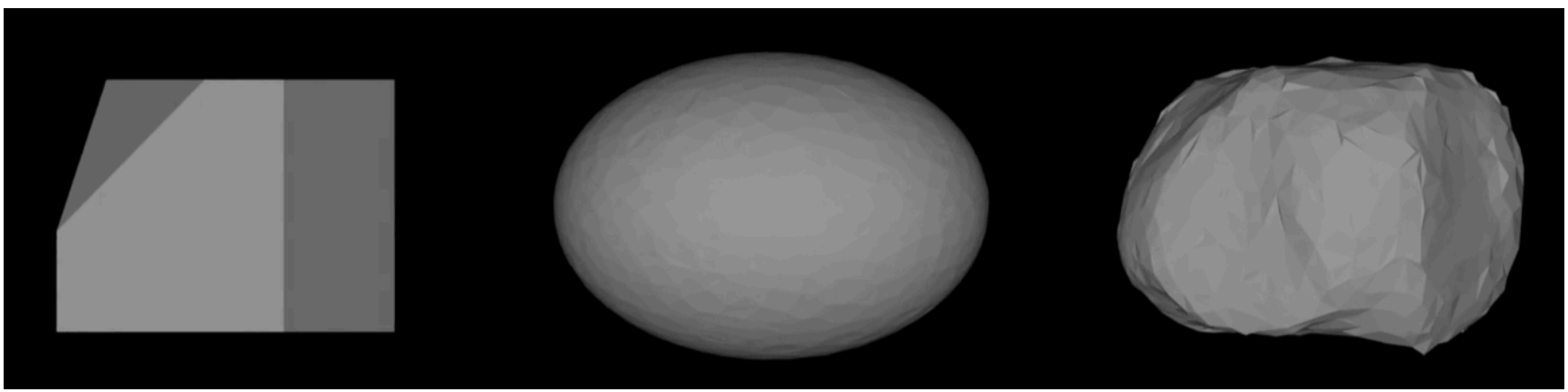
- Loss diverges quickly
- Likely due to imperfect forward model and excellent starting guess





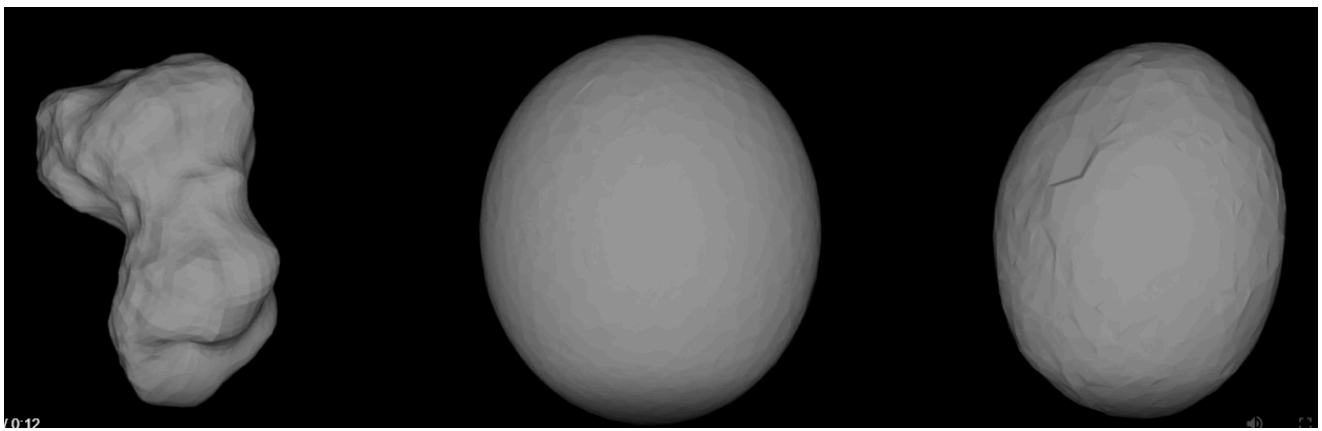
Reconstruction: Asteroid 2

- Honestly pretty good given naïve starting guess
- Improvements to consider:
  - Better forward model
  - Better starting guess (convex hull?)
  - Procedural mesh refinement



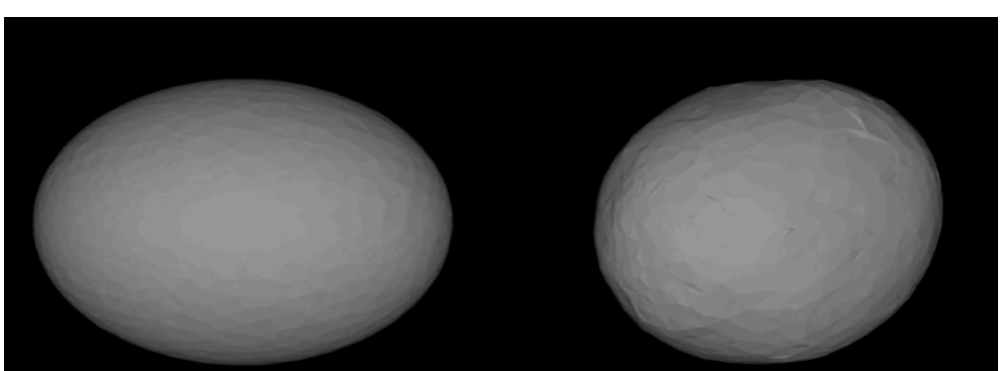
Reconstruction: Asteroid 3

- Introduces very slight non-convexity
- May need to improve optimization routine to explore non-convexity better
- Tilt is strangely to the wrong side?
  - Optimizer may have gotten stuck in local minimum



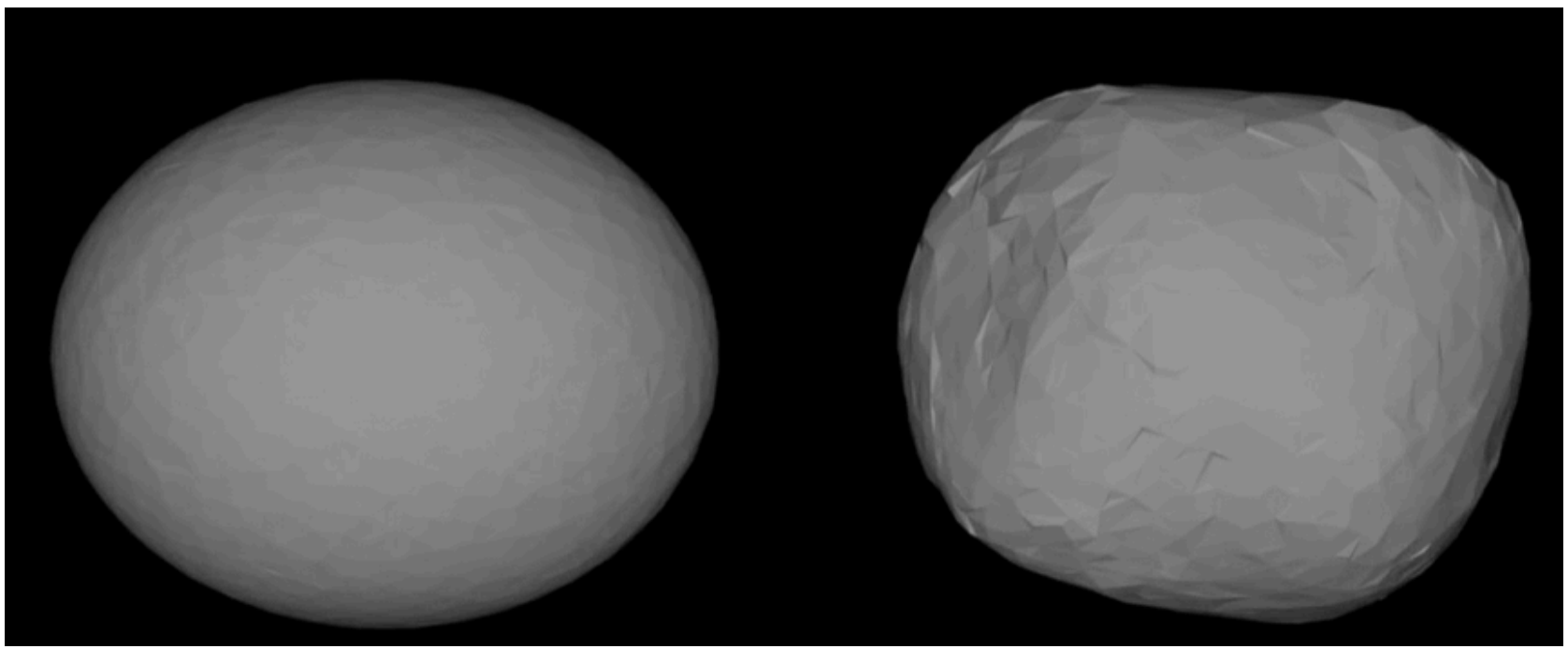
Reconstruction: Asteroid 4

- Successfully (?) rotates initial guess
  - A little hard to see in Figure 7 — easier in movie, **Live Demo?**
- We should explore different seeds and optimizations to see if we have any confidence in produced shape
- Actually quite nice lightcurve fit and loss, **Live Demo?**
- Optimize initial guess rapidly by optimizing for rotation of initial cylinder(s) only?
- Optimize initial guess by generating random meshes and picking the best one?



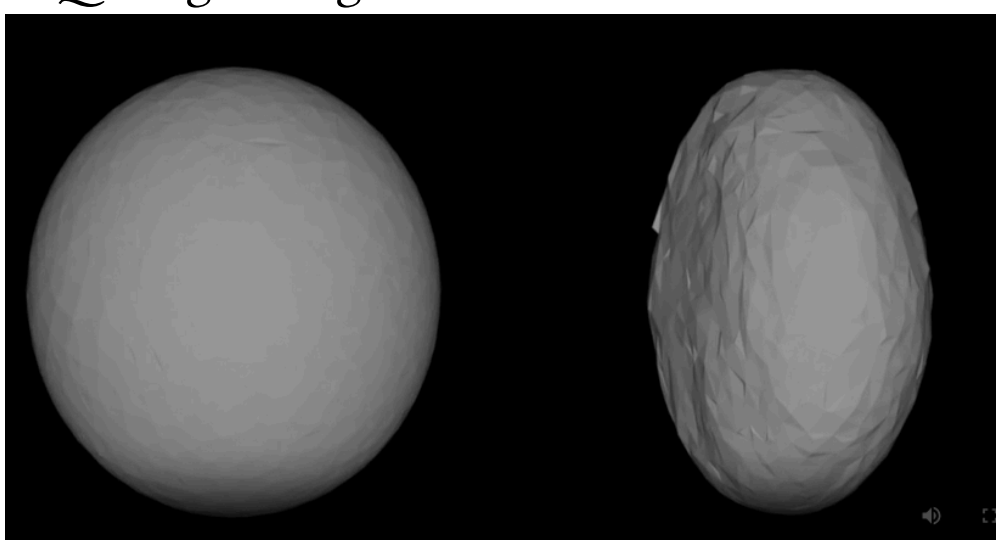
Reconstruction: Asteroid 5

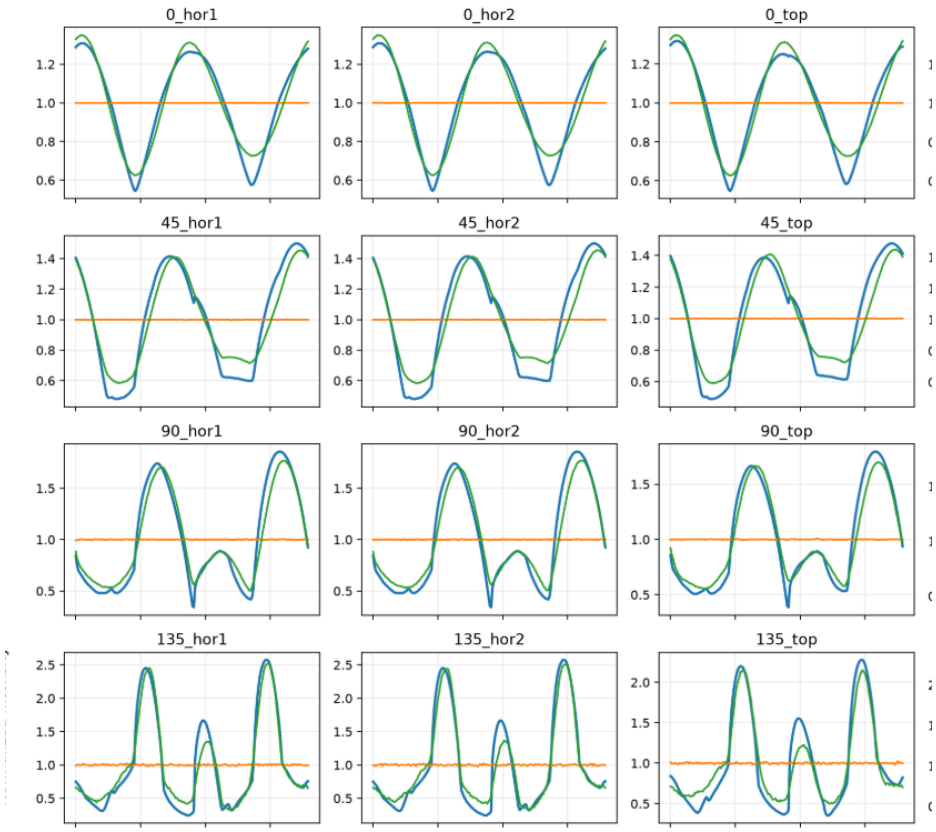
- At least has not exploded
- Reconstructions are a little ‘spiky’, maybe need procedural mesh normalization/ refinement?



Reconstruction: Asteroid 6

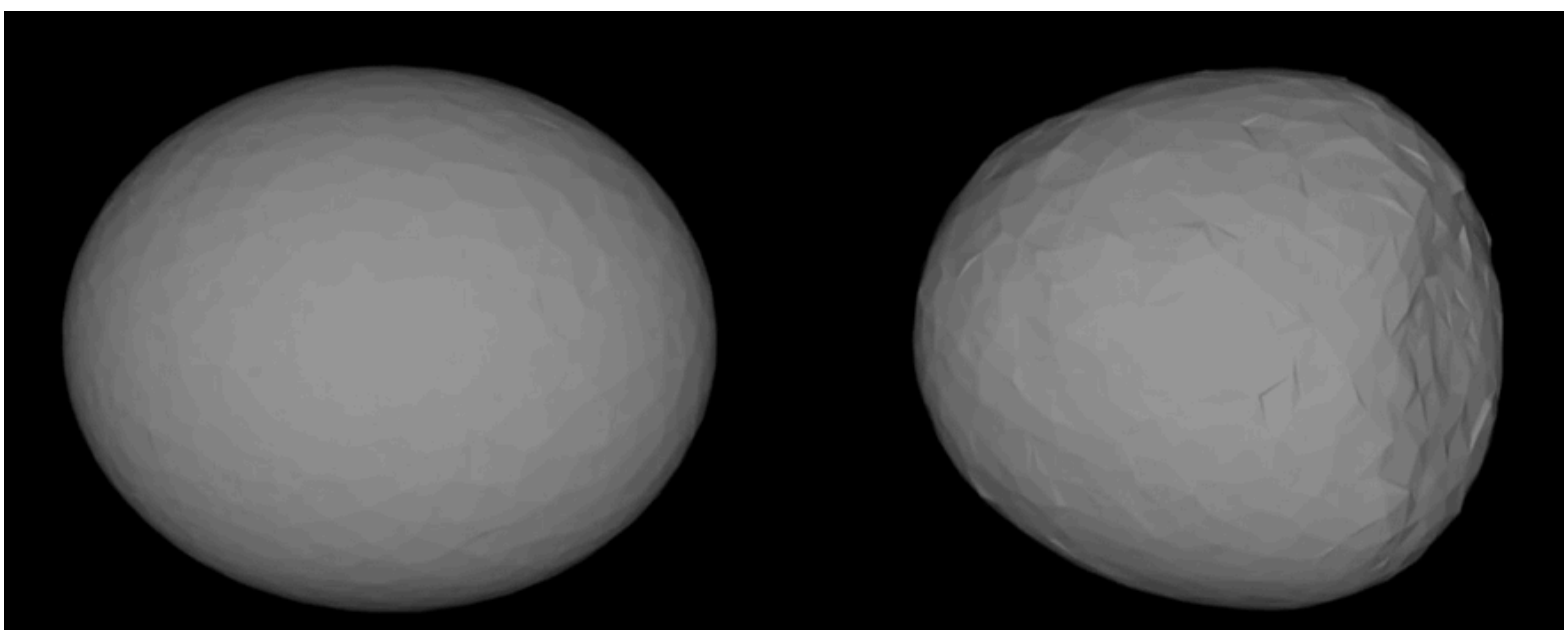
- Has a very flat face
- Again drawing many initial guesses from a prior distribution of archetypes may really help
  - Or if we can good convex hull?
- Quite good lightcurves!





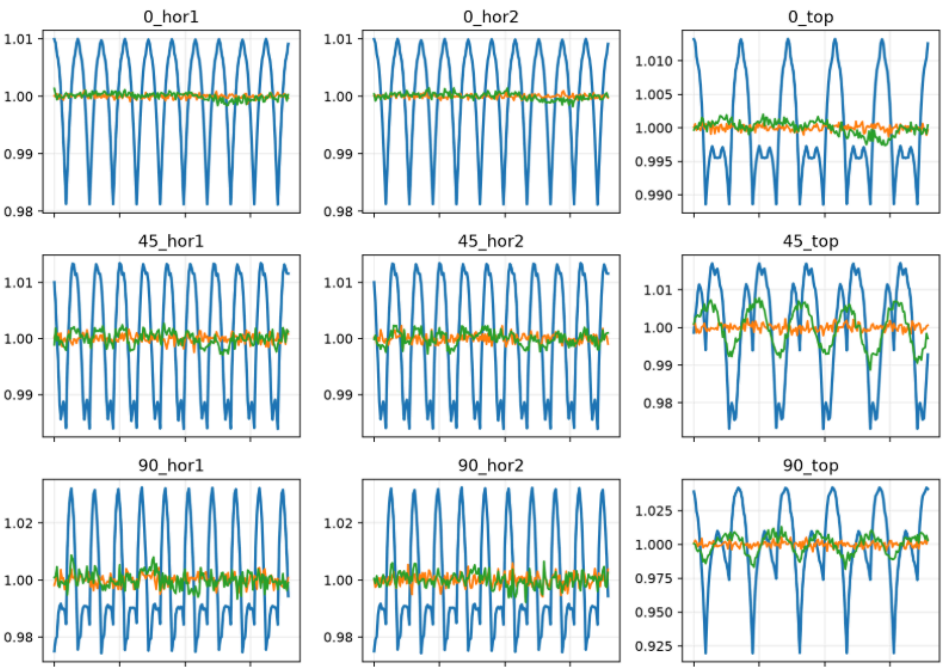
Reconstruction: Asteroid 7

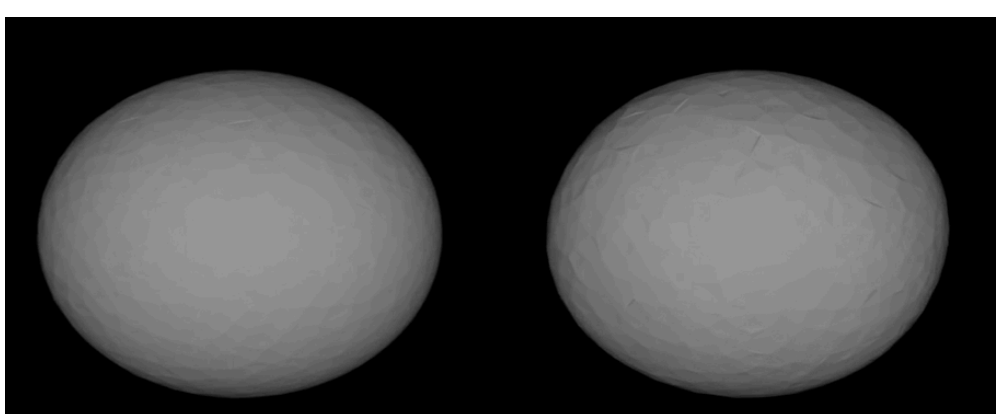
- I bet this has a hole it in or something awful like that
- We need initial guesses with different topologies, or some occasional refinement step that can make holes
- Very difficult since it will be highly non-differentiable to poke a hole in a mesh!



Reconstruction: Asteroid 8

- Very poor reconstruction
- From lightcurve: clearly extreme structure
  - Exploit lightcurve periodicity to make better starting guess
  - Can generally inform starting guess by lightcurve amplitudes and periodicities?





Reconstruction: Asteroid 9-10

- Still running when I finished up this presentation, but probably done now!
- Lets look! **Live Demo?**

Lightcurves

- Lightcurves tend to fit best for observation angles $\theta \in \{0°, 45°, 270°, 315°\}$, less so for $\theta \in \{135°, 225°\}$
- Should be improved with better forward model
  - Better scattering model
  - Better summation/reduction method? (thresholding?)

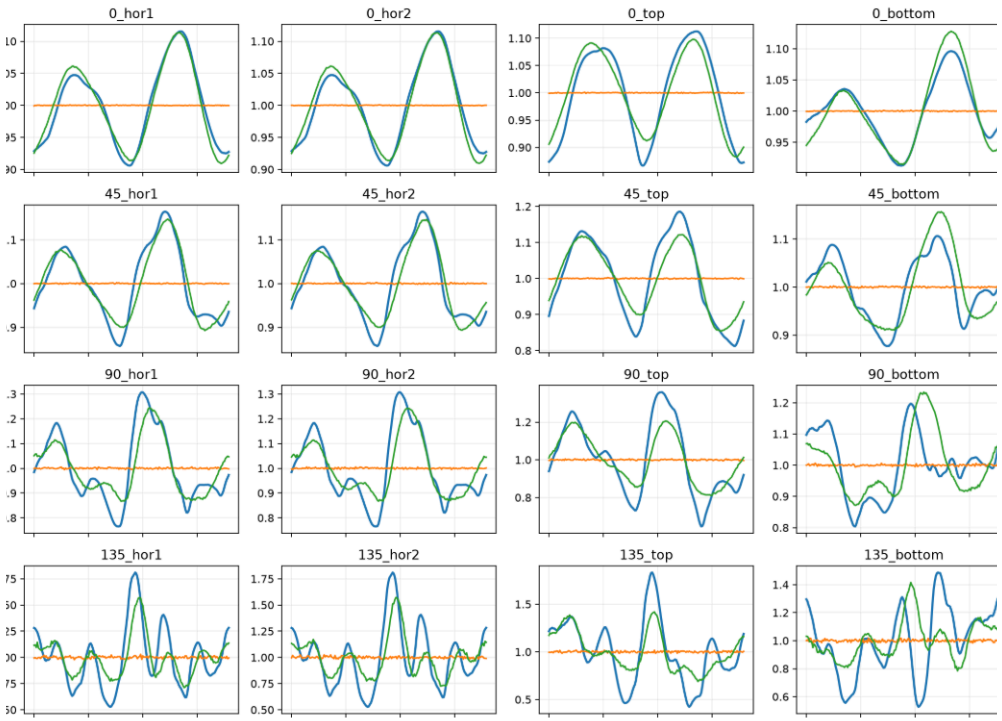


Figure 14: Lightcurve comparison for asteroid 3.

Summary/Coming up

- Try to improve forward model parameters by comparing known model meshes to their forward projections
- Make `ForwardModel` take a reduction method instead of hardcoding sum for lightcurve
- Maybe see if we can get speed-up with `u16` or `u8` instead of `u32`/`f32`?
- Making an `optimize` module under `hac.mitsuba` for common optimization routines
- Procedural mesh refinement?
- Should use all phases (360 for Blender, 842 for camera)
  - In current setup problem is actually over-determined, IIRC
- Should use camera data instead of only Blender data
- Speed up optimization (maybe some DrJit hackery? JIT Caching?)
- Asked HAC organisers whether Blender data has been corrected or not
  - No response yet

Final remarks

- This method probable needs a good starting guess to work well
- Can we use Mitsuba forward with a convex inversion method?
- Would be keen to meet up with people in DTU Library / 307 for a half-day?
- Mainly note to self:
  - Start putting renderings, logs, etc in `/work3/projects/HAC-DTU/workspaces/<USER>`
  - Currently can't be seen by others
  - Most of these don't belong in `git`
  - Output from reconstructions shown can be found there (if I remembered to do it?)